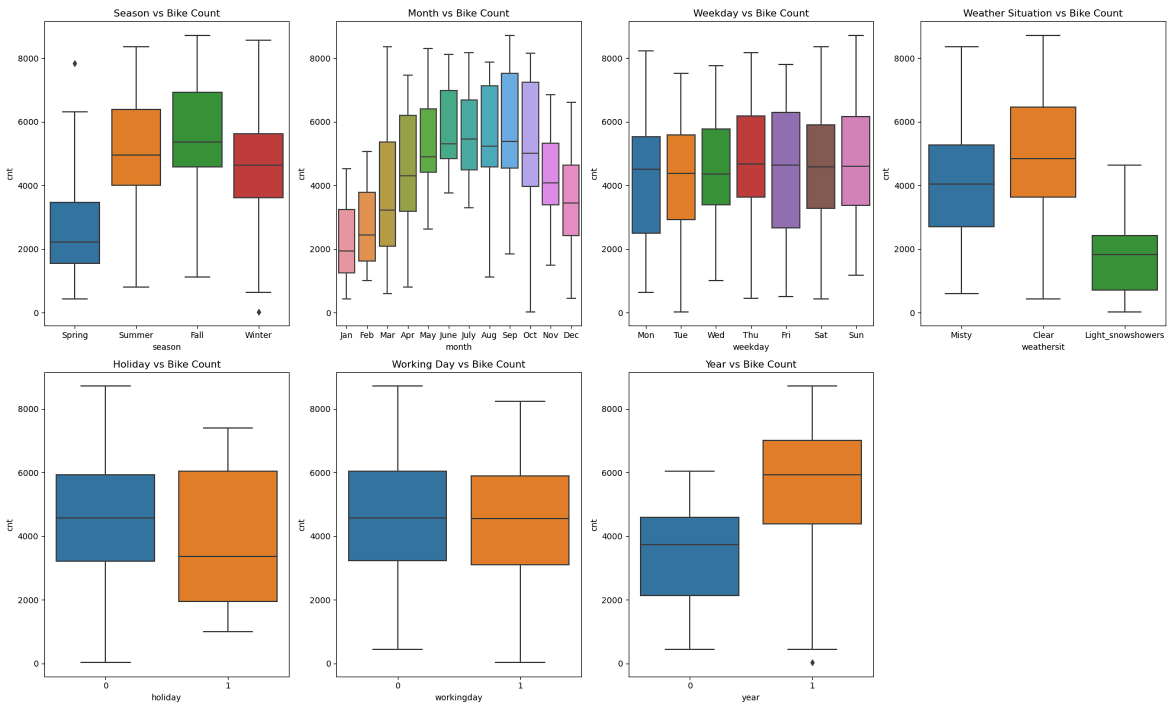
Assignment-based Subjective Questions

1. From your analysis of the categorical variables from the dataset, what could you infer about their effect on the dependent variable? (3 marks)



Based on the analysis of categorical columns using boxplots and bar plots, several insights can be drawn:

- The **fall season** appears to have had the highest number of bookings, with a significant increase in bookings observed across all seasons from 2018 to 2019.

- **May, June, July, August, September, and October** saw the highest volume of bookings, with a noticeable upward trend from the start of the year peaking around mid-year, followed by a decline towards the year-end.
- **Clear weather** was the most favorable for bookings, which aligns with common expectations.
- **Thursdays, Fridays, Saturdays, and Sundays** experienced higher booking activity compared to the earlier days of the week.
- Bookings tended to be lower on **non-holiday** days, which is understandable as people likely prefer to stay home and enjoy time with family during holidays.
- The number of bookings on **working days** and **non-working days** were fairly consistent, showing no significant difference.
- **2019** saw a substantial increase in bookings compared to 2018, reflecting positive business growth.

2. Why is it important to use **drop_first=True** during dummy variable creation? (2 mark)

Using drop_first=True during dummy variable creation is important because it helps to avoid the creation of redundant columns, thus reducing the risk of multicollinearity among dummy variables. When you have a categorical variable with **n** levels, only **n-1** columns are needed to represent the dummy variables.

For example, if a categorical column has 3 values (A, B, and C), creating dummy variables for all three would result in three columns. However, if a record is neither A nor B, it must be C by default, so there's no need for a separate column to represent C. By setting drop_first=True, the first category (A, for example) is dropped, and we only need two columns to represent the remaining categories (B and C). This reduces redundancy and the correlation between dummy variables.

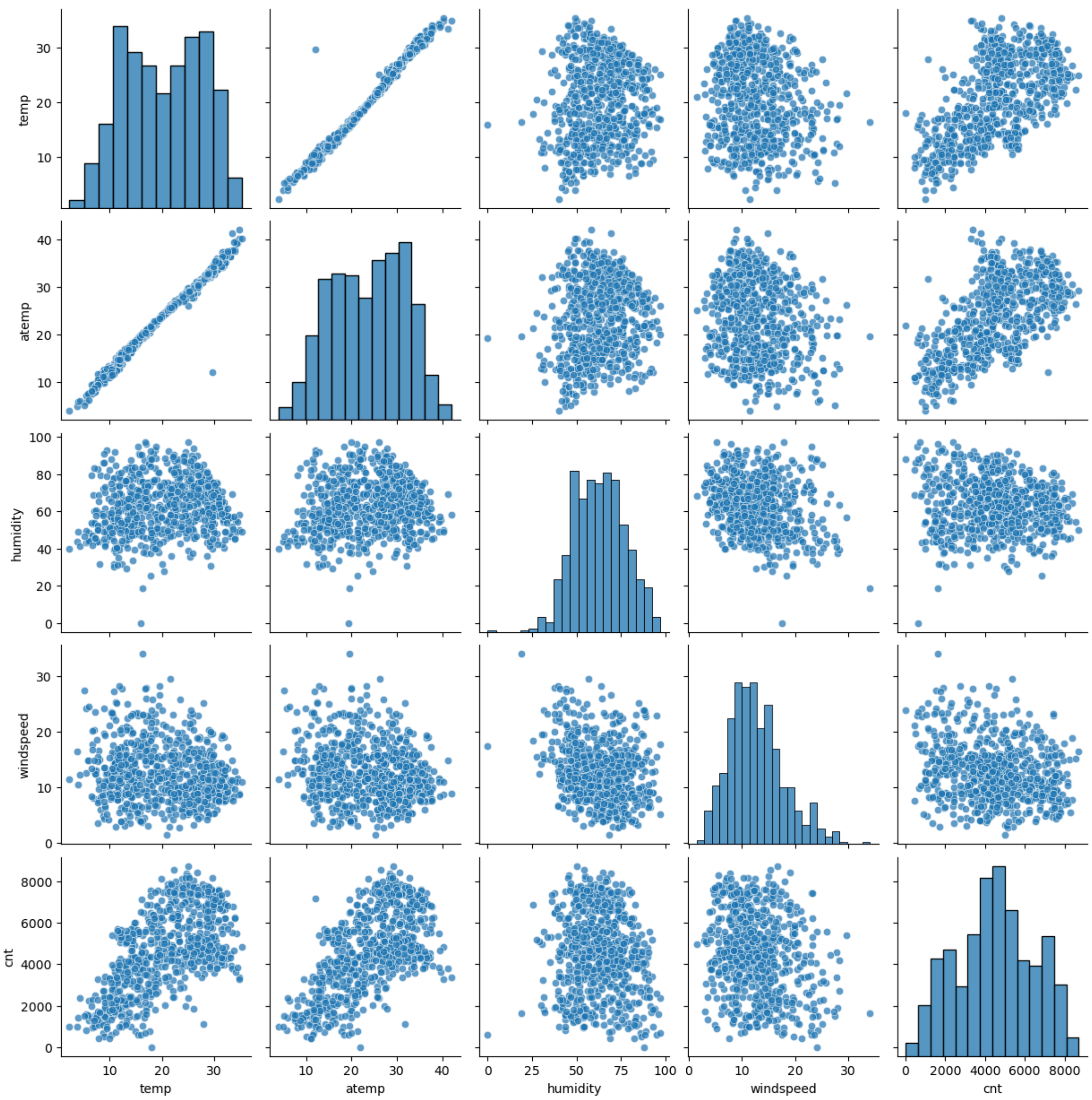
Syntax: drop_first: bool, default False — specifies whether to drop the first category (which is considered the reference category) to create **n-1** dummy variables.

In summary, drop_first=True ensures that the dummy variable creation is efficient and avoids unnecessary columns, which also helps prevent multicollinearity issues in regression models.

3. Looking at the pair-plot among the numerical variables, which one has the highest correlation with the target variable? (1 mark)

"temp" and "atemp" are the two numerical variables which are highly correlated with the target variable (cnt).

Pairplot of Numerical Features



4. How did you validate the assumptions of Linear Regression after building the model on the training set? (3 marks)

To validate the assumptions of linear regression after building the model on the training set, the following steps were taken:

1. **Linearity Assumption**:
   - **Visualized relationships** between the dependent variable and independent variables using scatter plots and Component + Residual (CCPR) plots to ensure a linear relationship between predictors and the target variable. If a non-linear pattern was observed, it would indicate a violation of the linearity assumption.
2. **Normality of Errors**:
   - **Histogram of residuals** was plotted to check if the error terms (residuals) follow a normal distribution. The assumption of normality is crucial for valid statistical inference. Additionally, a **Q-Q plot** was used to visually assess whether the residuals deviate from a normal distribution.
3. **Homoscedasticity**:
   - Plotted the **residuals vs. fitted values** to check for homoscedasticity (constant variance of errors). A random scatter pattern would confirm this assumption, while any discernible pattern (such as a funnel shape) would suggest heteroscedasticity.
4. **Independence of Errors**:
   - The assumption of **independence** was checked using residual plots and, where

applicable, Durbin-Watson test statistics. A value close to 2 indicates no autocorrelation in residuals.

5. **Multicollinearity**:
    - Calculated **Variance Inflation Factors (VIF)** to check for multicollinearity among predictors. A VIF greater than 5 (or 10, depending on the threshold) suggests high multicollinearity, which could skew model results.

By validating these assumptions, we can confirm the reliability of the linear regression model and its predictive power.

5. Based on the final model, which are the top 3 features contributing significantly towards explaining the demand of the shared bikes? (2 marks)

The top 3 significant features are:

1. temp : 0.477737
2. year : 0.234132
3. weathersit_Light_snowshowers : 0.285031

General Subjective Questions

1. Explain the linear regression algorithm in detail. (4 marks)

Linear Regression is a statistical method used for predicting the value of a dependent variable (also known as the target or outcome) based on one or more independent variables (also known as predictors or features). The relationship between the dependent and independent variables is modeled using a straight line, which is why it is called "linear" regression.

The **objective** of linear regression is to find the best-fitting line (or hyperplane in the case of multiple variables) that minimizes the difference between the observed values and the predicted values of the dependent variable.

1. Simple Linear Regression:

In the case of **simple linear regression**, we have only one independent variable and one dependent variable. The model follows the equation:

$y=\beta 0+\beta 1x+\epsilon$

Where:

- Y is the dependent variable (target).
- x is the independent variable (predictor).
- β0 is the intercept (the value of y when x=0).
- β1 is the slope (the change in y for a one-unit change in x).
- ϵ is the error term, representing the difference between the actual and predicted values.

The goal is to find the values of β0 and β1 that minimize the sum of squared errors (differences between observed and predicted values).

2. Multiple Linear Regression:

When there are multiple independent variables, the model becomes **multiple linear regression**. The equation is generalized as:

$y=\beta_0+\beta_1x_1+\beta_2x_2+...+\beta_nx_n+\epsilon$

Where:

- y is the dependent variable.
- $X_1,x_2,...,x_n$ are the independent variables.
- $B_0,\beta_1,...,\beta_n$ are the coefficients that the model needs to learn.
- ϵ is the error term.

The goal is still to find the optimal values of the coefficients $\beta_0,\beta_1,...,\beta_n$ that minimize the sum of squared residuals.

3. Steps in Linear Regression:

1. **Data Collection:** The first step is to collect data for the dependent and independent variables.
2. **Model Assumptions:** Linear regression assumes:
    - The relationship between variables is linear.
    - The residuals (errors) are normally distributed.
    - Homoscedasticity (constant variance of the residuals).
    - Independence of the observations.
3. **Training the Model:** Linear regression is trained by fitting the model to the data using optimization techniques, usually the **Ordinary Least Squares (OLS)** method. This method minimizes the sum of squared differences between the observed values and the predicted values.
4. **Prediction:** Once the model is trained, it can be used to make predictions by applying the learned coefficients to new data.
5. **Evaluation:** The model is evaluated using metrics such as **R-squared**, **Adjusted R-squared**,

Mean Squared Error (MSE), and **Root Mean Squared Error (RMSE)**.

4. Optimization (Ordinary Least Squares):

The most commonly used method to estimate the coefficients in linear regression is **Ordinary Least Squares (OLS)**. This method works by minimizing the **cost function**, which is the sum of squared residuals (errors) between the predicted values and the actual values.

5. Model Evaluation:

To assess the performance of the model, we commonly use the following metrics:

- **R-squared (R^2):** Represents the proportion of the variance in the dependent variable that is predictable from the independent variables. It is between 0 and 1, where 1 indicates a perfect fit.
- **Adjusted R-squared:** Adjusts R^2 for the number of predictors in the model, helping to avoid overfitting.
- **Mean Squared Error (MSE):** Measures the average squared difference between actual and predicted values.
- **Root Mean Squared Error (RMSE):** The square root of MSE, providing the error in the same units as the dependent variable.

Conclusion:

Linear regression is a powerful and widely used algorithm that helps in modeling the relationship between variables. It's simple to implement and understand, making it a good starting point for predictive modeling tasks. However, it assumes linearity and may not work well with non-linear relationships or if the assumptions are violated.

2. Explain the Anscombe's quartet in detail. (3 marks)

Anscombe's Quartet: An Explanation

Anscombe's Quartet is a set of four datasets that have nearly identical simple descriptive statistics but appear very different when graphed. The datasets were constructed by the statistician **Francis Anscombe** in 1973 to demonstrate the importance of graphing data before analyzing it, as relying solely on descriptive statistics can be misleading.

The Four Datasets:

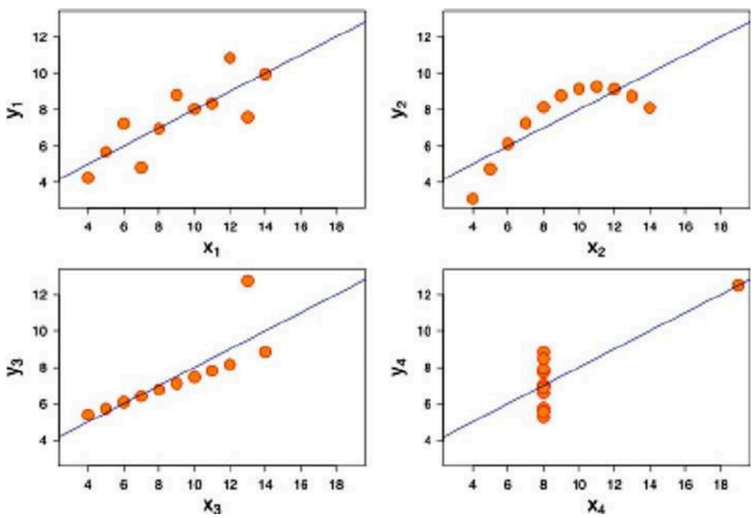
Anscombe's quartet consists of four datasets, each containing 11 data points. Here's a summary of their key features:

Dataset	Mean of X	Mean of Y	Variance of X	Variance of Y	Correlation	Regression Line (Y = a + bX)
I	9	7.5	11	3.5	0.82	Y = 3 + 0.5X
II	9	7.5	11	3.5	0.82	Y = 3 + 0.5X
III	9	7.5	11	3.5	0.82	Y = 3 + 0.5X

IV	9	7.5	11	3.5	0.82	Y = 3 + 0.5X

Key Points:

1. **Identical Descriptive Statistics:** All four datasets have the same:
    - Mean of X = 9
    - Mean of Y = 7.5
    - Variance of X = 11
    - Variance of Y = 3.5
    - Correlation between X and Y = 0.82
    - The regression line of the form Y = 3 + 0.5X
2. **Divergence in Graphical Representation:**
    - **Dataset I**: Appears to show a linear relationship between X and Y. This dataset aligns with the regression line well, showing a classic positive correlation.
    - **Dataset II**: Also shows a linear relationship, similar to Dataset I. However, the points are scattered around the regression line, and there are no outliers.
    - **Dataset III**: Here, the data points follow a curve. Even though the regression line fits the data poorly, the statistical summary (mean, variance, correlation) looks similar to the other datasets.
    - **Dataset IV**: This dataset contains a significant outlier. The outlier dramatically changes the graph and influences the regression line, but the summary statistics still match the others.

- 

Purpose and Lessons from Anscombe's Quartet:

- **Graphical Visualization is Crucial:** Anscombe's quartet illustrates that even if the descriptive statistics are identical, the data can behave in completely different ways. Graphing the data reveals important insights about the underlying relationship and potential issues, such as outliers or non-linearity, which statistical summary measures cannot capture.
- **Statistical Summary Alone Can Be Misleading:** Although the datasets in Anscombe's quartet have the same correlation and regression line, the plots reveal that relying solely on these statistics would lead to incorrect interpretations. The non-linearity and outliers in some datasets would have been overlooked without visualization.

Conclusion:

Anscombe's quartet highlights the importance of examining data visually before drawing conclusions. It reinforces that statistical analysis, though useful, should always be complemented with data visualization to avoid misleading interpretations and ensure robust data analysis.

3. What is Pearson's R? (3 marks)

Pearson's R (Pearson Correlation Coefficient):

Pearson's R, also known as the **Pearson correlation coefficient**, is a statistical measure that quantifies the strength and direction of the linear relationship between two continuous variables. It is represented by the symbol **r** and ranges from **-1 to +1**.

Formula

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

r = correlation coefficient

x_i = values of the x-variable in a sample

$\bar{x}$ = mean of the values of the x-variable

y_i = values of the y-variable in a sample

$\bar{y}$ = mean of the values of the y-variable

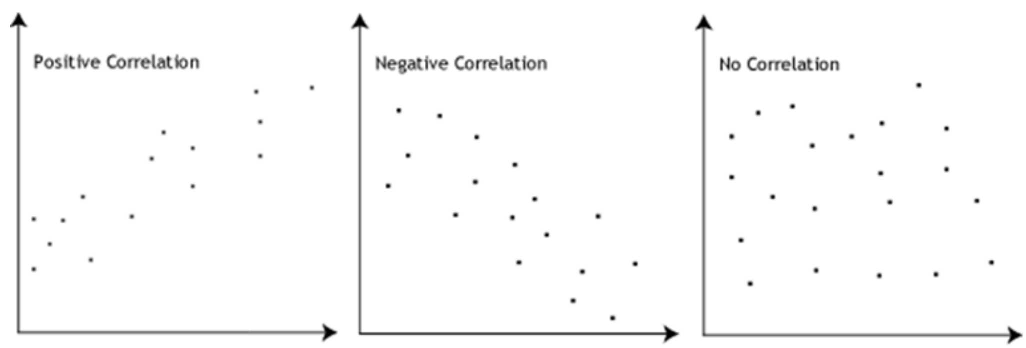
Interpretation:

- **r = +1**: A perfect positive linear relationship. As x increases, yy increases in a perfectly linear fashion.
- **r = -1**: A perfect negative linear relationship. As x increases, yy decreases in a perfectly linear fashion.
- **r = 0**: No linear relationship between xx and yy. There may still be some non-linear relationship, but the variables are uncorrelated in a linear sense.
- **r > 0**: A positive correlation, meaning that as one variable increases, the other tends to increase as well.
- **r < 0**: A negative correlation, meaning that as one variable increases, the other tends to decrease.

Strength of the Relationship:

- **0.1 to 0.3 (or -0.1 to -0.3)**: Weak positive (or negative) correlation.
- **0.3 to 0.5 (or -0.3 to -0.5)**: Moderate positive (or negative) correlation.
- **0.5 to 1.0 (or -0.5 to -1.0)**: Strong positive (or negative) correlation.

Significance of Pearson's R:



- **Positive Correlation**: If rr is positive, there is a tendency that as one variable increases, the other will also increase.
- **Negative Correlation**: If rr is negative, there is a tendency that as one variable increases, the other will decrease.
- **No Correlation**: If rr is close to 0, there is no clear linear relationship between the variables.

Assumptions:

- Both variables are continuous.
- The relationship between the variables is linear.
- The data is homoscedastic (i.e., the variance of one variable is constant across the values of the other variable).
- The data points are independent.

In conclusion, **Pearson's R** is a fundamental tool for understanding the strength and direction of the linear relationship between two variables, helping in various fields of statistics and data analysis to assess correlations.

4. What is scaling? Why is scaling performed? What is the difference between normalized scaling and standardized scaling? (3 marks)

Feature Scaling is a technique used to standardize the independent features in a dataset by transforming them into a fixed range. This is typically performed during data preprocessing to handle features with varying magnitudes, values, or units. Without feature scaling, machine learning algorithms may assign higher weights to features with larger values and lower weights to those with smaller values, regardless of their actual units.

For example, without feature scaling, an algorithm may interpret 3000 meters as larger than 5 kilometers, which is incorrect. In such cases, feature scaling helps bring all feature values to a similar magnitude, preventing such issues and leading to more accurate predictions.

S.NO.	**Normalized Scaling**	**Standardized Scaling**
1.	Uses the minimum and maximum values of features for scaling	Uses the mean and standard deviation for scaling.
2.	Applied when features have different scales.	Applied when we need to achieve zero mean and unit standard deviation.
3.	Scales values within a specific range, typically [0, 1] or [-1, 1].	Does not limit values to a specific range.
4.	Sensitive to outliers.	Less affected by outliers.
5.	Scikit-Learn provides the MinMaxScaler for normalization.	Scikit-Learn provides the StandardScaler for standardization.

5. You might have observed that sometimes the value of VIF is infinite. Why does this happen? (3 marks)

The Variance Inflation Factor (VIF) can become infinite in the following situations:

1. **Perfect Multicollinearity**: VIF becomes infinite when there is perfect multicollinearity

between one predictor variable and one or more other predictor variables in the model. This means that one predictor can be expressed as an exact linear combination of others, leading to a situation where the matrix used for calculating the regression coefficients is singular and cannot be inverted. In such cases, the variance of the coefficient estimates becomes undefined, resulting in an infinite VIF value.

2. **Redundancy Among Variables**: If two or more predictor variables are perfectly correlated (i.e., their correlation is 1 or -1), the VIF for these variables will approach infinity. In this case, it becomes impossible to distinguish their individual contributions to the model because they are essentially providing the same information.

3. **Omitted Variables**: VIF can also become infinite if there is an important predictor variable that has been omitted from the model. If the model doesn't include this variable, it may artificially inflate the variance of the remaining predictors, leading to high or infinite VIF values.

In these cases, it's recommended to either remove one of the highly correlated predictors or use techniques like Principal Component Analysis (PCA) to reduce multicollinearity.

6. What is a Q-Q plot? Explain the use and importance of a Q-Q plot in linear regression. (3 marks)

A **Q-Q plot (Quantile-Quantile plot)** is a graphical tool used to assess whether a dataset follows a certain theoretical distribution, most commonly the normal distribution. It plots the quantiles of the dataset against the quantiles of the chosen reference distribution (e.g., normal distribution). If the data follows the reference distribution, the points in the Q-Q plot will approximately lie on a straight line.

Use and Importance of a Q-Q Plot in Linear Regression:

1. **Normality of Residuals**:
   - In linear regression, one of the key assumptions is that the residuals (errors) of the model should be normally distributed. A Q-Q plot is an effective way to visually inspect the distribution of residuals. If the residuals are normally distributed, the points in the Q-Q plot will lie along the straight diagonal line.
2. **Detecting Deviations from Normality**:
   - By examining the Q-Q plot, you can detect deviations from normality. For example, if the residuals exhibit heavy tails (i.e., points deviate away from the line at the ends), this might indicate a non-normal distribution such as a heavy-tailed distribution (e.g., t-distribution).
3. **Model Assumptions**:
   - The Q-Q plot helps validate the assumption of normality of errors. If the residuals are not normally distributed, the results of hypothesis tests (like t-tests) in linear regression might not be valid. In such cases, transformations of the dependent variable or the use of non-parametric models may be necessary.
4. **Outlier Detection**:
   - The Q-Q plot also helps in detecting outliers. Points that deviate significantly from the line suggest the presence of outliers in the data, which may influence the regression results. Identifying these outliers helps improve model accuracy and interpretability.

In summary, a Q-Q plot is a vital diagnostic tool in linear regression to ensure that the residuals follow the assumption of normality, which underpins many statistical tests and conclusions.